

Weekly CT-02 Answer Sheet

Cambridge International AS and A Level Computer Science 9618
Networks including the internet

Maximum Marks	30
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Marking note: Award marks for equivalent wording that communicates the same technical point.

1. LAN benefits and characteristics [4]

Benefits: award 1 mark each, maximum 2:

- Allows sharing of files/data or communication between devices. 1
- Allows sharing of resources, e.g. hardware or software/applications. 1
- Allows central management, e.g. backup or security. 1

Characteristics: award 1 mark each, maximum 2:

- Covers a small geographical area. 1
- The infrastructure is privately owned / not controlled by external organisations. 1

2. Network devices and transmission media [7]

(a) 1 mark for each completed name or description:

- Router - receives and sends data between two networks operating on the same protocol. 1
- Wireless Network Interface Card (WNIC) - hardware component that allows a device to connect to a wireless network; provides a MAC address to identify the device on the wireless network. 1
- Repeater - restores the digital signal so it can be transmitted over greater distances. 1
- Wireless Access Point (WAP) - hardware component that provides radio communication from the central device to nodes on the network, and vice versa. 1

(b) Award 1 mark for each valid difference, maximum 3. Accept, for example:

- Fibre-optic data is transmitted using light; copper cable uses electrical signals. 1
- Fibre optic has higher bandwidth / higher transmission rates than copper cable. 1
- Fibre optic has less risk of noise/interference than copper cable. 1
- Fibre optic can be used over longer distances before repeaters are needed. 1
- Fibre optic is much more difficult to hack into than copper cable. 1
- Fibre optic is more prone to damage than copper cable. 1

3. Cloud computing

[6]

- (a) Accessing a service/files/software on a remote server. 1
 - (b) Public cloud: computing services offered by a third-party provider over the public Internet; open/available to anyone with the appropriate equipment/software/credentials. 1
 - Private cloud: computing services offered over the Internet or a private internal network and available only to selected users / the organisation, not the general public. 1
- (c) Benefits: award 1 mark each, maximum 2. Accept, for example:
- Can be accessed anywhere with Internet access. 1
 - No need to perform backups / no need to buy specific software or hardware / documents can be shared easily / multiple people can work on the same document. 1
 - Drawback: award 1 mark. Accept, for example: cannot access without Internet; reliant on a third party for backup/security; cannot access if the server goes down. 1

4. CSMA/CD

[4]

Award 1 mark per valid point, maximum 4:

- CSMA/CD is a protocol used to detect and prevent collisions in a bus topology. 1
- Before transmitting, a device checks whether the communication channel is busy. 1
- If the channel is busy the device waits; if it is free, the data is sent. 1
- Because more than one computer shares the same transmission medium, two workstations can start transmitting at the same time, causing a collision. 1
- If a collision is detected, transmission is aborted / a jamming signal is transmitted. 1
- Both devices wait for a different random amount of time and then try again. 1

5. IPv4 and IPv6

[5]

- IPv4 contains 4 groups of digits. 1
- Each IPv4 group is represented in 8 bits. 1
- IPv6 contains 8 groups of digits. 1
- Each IPv6 group is represented in 16 bits. 1
- Multiple groups containing only zeros can be replaced with :: (double colon). 1

6. Router functions

[3]

Award 1 mark per valid point, maximum 3. Accept, for example:

- Receives packets from the Internet / an external network. 1
- Analyses the destination IP address of each packet and forwards the packet towards its destination. 1
- Uses and maintains/updates a routing table; finds the most efficient path to the destination. 1
- Can allocate private IP addresses / perform Network Address Translation (NAT) / implement a firewall / change packet format for the next network. 1

7. Bonus: study time

[0.5]

- At least 2 hours of mindful study is expected. 0.5

8. Bonus: participation

[0.5]

- Thank you again. 0.5

Total

30 marks

(29 marks + 1 bonus mark)